

Accelerating Lifelong MAPF Optimization via Uncertainty-Based Surrogate-Assisted Evolution Strategies

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Abstract—Lifelong Multi-Agent Path Finding (LMAPF) is computationally demanding. Optimizing guidance graphs with evolutionary algorithms such as CMA-ES typically requires hundreds of thousands of expensive multi-agent simulations. This paper introduces an uncertainty-aware surrogate-assisted approach that replaces the majority of those simulations with predictions from an ensemble of Multi-Layer Perceptrons. An Upper Confidence Bound (UCB) selection mechanism balances exploitation of high-predicted-fitness candidates with exploration of uncertain search-space regions, preventing surrogate exploitation as the CMA-ES distribution shifts into unexplored regions during evolution. The surrogate reaches the same throughput quality as the full-budget baseline using 2.81–2.92x fewer simulations across two random seeds (49k vs. 137k simulations to first reach the same throughput threshold on the primary run). A gradient refinement experiment provides direct evidence of surrogate exploitation and confirms that the conservative UCB design guards against it. A diagnostic analysis using full-leakage, oracle, and noise-bound tests identifies high inter-emitter population diversity, quantified as intraclass correlation (ICC), as a necessary condition for surrogate effectiveness. A single-emitter baseline without any surrogate reaches the same quality threshold at 3.30x fewer simulations than the vanilla multi-emitter run, confirming that emitter focusing rather than surrogate ranking is the primary efficiency mechanism.

Index Terms—lifelong multi-agent path finding, surrogate-assisted optimization, CMA-ES, uncertainty quantification, Upper Confidence Bound, evolutionary computation, intraclass correlation

I. INTRODUCTION

Lifelong Multi-Agent Path Finding is central to modern automated logistics. It manages the robot fleets in fulfillment centers such as those operated by Amazon and Ocado. Unlike traditional path finding, where agents stop after reaching a single destination, LMAPF is a continuous process: agents receive new goals immediately upon completing their current task. The standard approach to this problem is Guidance Graph Optimization (GGO), introduced by Zhang et al. [1], in which a Covariance Matrix Adaptation Evolution Strategy (CMA-ES) learns a set of flow weights that overlay the warehouse

map. These weights guide decentralized agents toward high-throughput routes and reduce congestion.

A key bottleneck limits the scalability of GGO. To evaluate a single candidate set of flow weights, the optimizer must run multiple complete multi-agent simulations, each requiring hundreds of timesteps to produce a stable throughput measurement. A standard 300-generation run requires 150,000 simulations and roughly 9.5 hours of wall-clock time on an 8-core machine. This makes the method too slow for dynamic environments where warehouse layouts or goal distributions change frequently, and limits the number of configurations that can be tested during initial deployment. The online extension of GGO explicitly identifies surrogate-assisted optimization as a direction for future work [2].

This paper addresses the problem by integrating a machine learning surrogate into the CMA-ES loop. The surrogate is trained on observed (solution, throughput) pairs and used to pre-screen candidate solutions before simulation. Only the most promising candidates and those in unexplored regions are passed to the simulator. Three research questions guide the work. First, how can a surrogate model be integrated into the CMA-ES framework for LMAPF throughput prediction? Second, how much does surrogate-assisted pre-screening reduce computation time relative to the baseline? Third, how does uncertainty tracking affect surrogate stability and prevent exploitation compared to a point-estimate model? Beyond these questions, the analysis examines what structural conditions make surrogate pre-screening feasible at all, identifying population diversity as a necessary precondition whose absence renders any surrogate ineffective regardless of model capacity.

The main contributions are as follows. We show that an ensemble of bootstrapped MLPs achieves a Spearman rank correlation of 0.78–0.94 in a realistic sliding-window test, sufficient for effective pre-screening. The surrogate reaches equivalent throughput to the vanilla baseline using 2.81x fewer simulations (49k vs. 137k simulations to first reach $\tau = 8.23$). A gradient refinement experiment predicted a +0.95 throughput gain but measured a –0.11 decrease in real simulation, giving direct evidence of surrogate exploitation. Finally, we extend the framework to the online GGO setting of

Zang et al. [2]. The surrogate reduces the required simulations from 10,000 to 3,520 in the online setting, a 2.84x reduction, reaching equivalent throughput quality at a 2.41x crossover speedup.

II. BACKGROUND AND RELATED WORK

A. Guidance Graph Optimization

Zhang et al. [1] introduced an automated method for generating traffic rules for robot warehouses. Given a warehouse floor plan as a graph $G = (V, E)$, the optimization assigns a scalar flow weight $w_e \in \mathbb{R}$ to each directed edge $e \in E$. A decentralized pathfinding planner uses these weights as directional biases. Throughput τ measures the mean number of tasks completed per timestep at steady state, averaged over a fixed number of agents and simulation steps. The baseline optimizer uses CMA-ES [8] with multiple independent emitters [12]. Each emitter maintains a multivariate normal distribution over the weight space; at each generation it proposes 20 candidate solutions, evaluates their throughput, and updates its distribution parameters to move toward higher-throughput regions.

B. Surrogate-Assisted Evolutionary Computation

Surrogate-assisted evolutionary computation reduces the number of expensive fitness evaluations by replacing some with predictions from a cheaper model. The standard approach evaluates only a subset of each generation using the true fitness function and fills the rest with surrogate predictions [3]. A known problem with this strategy is surrogate exploitation: the optimizer finds regions where the model overestimates fitness and the search is directed away from genuine improvements.

Uncertainty quantification offers a way to address this. By measuring model confidence, the algorithm can balance exploiting high-predicted-fitness regions against exploring uncertain ones. The Upper Confidence Bound criterion, from the multi-armed bandit literature [6], captures this trade-off as a single scalar score and has been applied to Bayesian optimization and surrogate-assisted search [7]. This work applies it to multi-emitter CMA-ES, where natural distribution shifts during evolution move the search into unexplored regions and exploration matters most. Zang et al. [2] extend the GGO framework to an online setting, optimizing a CNN policy that adapts edge weights in real time based on observed traffic. Their paper explicitly suggests surrogate-assisted optimization as a direction for future work, which this paper addresses in Section V-G.

III. METHODOLOGY

A. Infrastructure and Baseline

The warehouse environment is represented as a directed graph where each edge carries a flow weight; the full weight vector $\mathbf{w} \in \mathbb{R}^{|E|}$ is the parameter optimized by CMA-ES, with $|E| = 4,074$ (948 wait weights and 3,126 directed edge weights on the warehouse map). Throughput $\tau(\mathbf{w})$ is the mean number of tasks completed per timestep, averaged over $S = 5$ simulation seeds and $T = 1,000$ timesteps per seed; S is

increased to 100 for final robustness comparisons. Because each evaluation requires a full multi-agent simulation, τ is the main computational bottleneck.

The optimization pipeline consists of a Python wrapper around a C++ LMAPF simulator running inside a Docker container. Multiprocessing across 8 workers gives a $2.1\times$ speedup, reducing per-generation wall-clock time from roughly 5 minutes to roughly 2 minutes. The baseline uses a multi-emitter CMA-ES with 5 emitters, each proposing 20 candidates, giving 100 candidates per generation. Over 300 generations this totals 150,000 simulations (100 candidates $\times$ 5 seeds $\times$ 300 generations) and roughly 9.5 hours of wall-clock time.

B. Surrogate Architecture Selection

We ran an offline feasibility study comparing XGBoost, Convolutional Neural Networks (CNN), and Multi-Layer Perceptrons (MLP) using a sliding-window temporal test: the model is trained on generations $[0, g)$ and tested on generation g across 8 test points, matching the live-loop condition. XGBoost matched MLP accuracy but required 100–250 seconds per training run, comparable to a full-evaluation generation, which would cancel out pre-screening savings. The CNN showed no consistent benefit from spatial representation of the weight vector (ρ as low as 0.05 at generation 10). The MLP was therefore selected; its three-layer (256-128-64) architecture provides sufficient capacity at the 2,000–20,000 sample range of this setting, with model weights kept between generations and incremental fine-tuning on 100 new samples taking only 2–5 seconds.

C. Surrogate-Assisted CMA-ES Loop

The surrogate-assisted pipeline replaces a fraction of full evaluations with model predictions each generation. The loop runs in three modes, shown in Fig. 1.

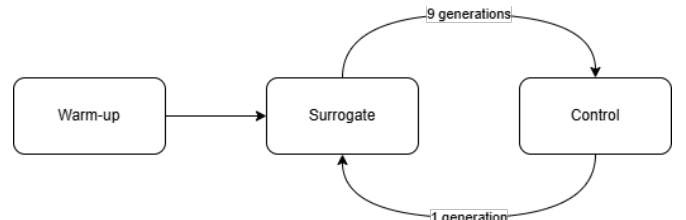


Fig. 1. The three-phase surrogate-assisted CMA-ES loop. The Surrogate Phase uses the UCB score $\text{score} = \mu + \lambda\sigma$ to select which of the 100 CMA-ES proposals receive full simulation.

During the *Warmup Phase* (generations 0–19), all 100 candidates receive full simulation to collect 2,000 training points before any pre-screening begins. Twenty warmup generations provide roughly double the minimum data volume at which the offline ensemble achieves reliable rank correlation (Section III-B), ensuring pre-screening does not begin on an undertrained model. During the *Evolution Control Phase*, triggered every 10 generations, all candidates again receive full evaluation and the ensemble is retrained on the full

history to prevent the model from going stale as the CMA-ES distribution shifts. The 10-generation interval limits full-evaluation overhead to roughly 10% of the simulation budget while providing a regular model refresh. During the *Surrogate Phase*, which covers most generations, the ensemble scores all 100 proposals and the top 20 by score are sent to the simulator. Surrogate predictions for the remaining 80 candidates are passed to the CMA-ES update as placeholder fitness values, and the ensemble is fine-tuned on the 20 new (solution, throughput) pairs. The 20-candidate limit targets a theoretical 5x speedup and is justified by the population structure: at ICC near 0.95 the surrogate reliably identifies which emitter a candidate belongs to, making it effective at separating the top proposals from the rest.

D. Ensemble UCB Acquisition Function

To estimate prediction uncertainty, we train 5 bootstrapped MLP instances on independently resampled subsets of the training data. Five members follows established practice for bootstrap ensembles in surrogate-assisted optimization [9], giving stable disagreement estimates without disproportionate training overhead. For a candidate solution $\mathbf{w}$, each model f_k produces a throughput prediction $\hat{\tau}_k(\mathbf{w})$. The ensemble mean and standard deviation are:

$$\mu(\mathbf{w}) = \frac{1}{5} \sum_{k=1}^5 \hat{\tau}_k(\mathbf{w}), \quad (1)$$

$$\sigma(\mathbf{w}) = \sqrt{\frac{1}{5} \sum_{k=1}^5 (\hat{\tau}_k(\mathbf{w}) - \mu(\mathbf{w}))^2}. \quad (2)$$

The UCB score used for candidate selection is:

$$\text{score}(\mathbf{w}) = \mu(\mathbf{w}) + \lambda \cdot \sigma(\mathbf{w}), \quad (3)$$

where $\lambda \geq 0$ controls the exploration weight. At $\lambda = 0$, candidates are ranked by mean prediction alone. As λ increases, candidates in regions with high ensemble disagreement receive a score boost and are prioritized for simulation. We test $\lambda \in \{0, 1.0, 2.0\}$ in Section V; the ablation selects $\lambda = 1.0$ as the final value.

This interacts directly with the natural distribution shifts that occur during CMA-ES evolution. As the population moves through the $|E|$ -dimensional weight space, the ensemble regularly encounters candidate regions it has not previously seen, and ensemble disagreement σ is large for those candidates. The UCB score then promotes them for simulation regardless of their mean prediction. Once the ensemble collects training data for the new region, σ drops over the next few control generations. In well-explored regions $\sigma \approx 0$, so the UCB score reduces to the mean prediction.

IV. EXPERIMENTAL SETUP

The 300-generation budget, 5-emitter configuration, and 20-candidates-per-emitter structure follow Zhang et al. [1] to ensure a comparable baseline. The baseline ran for 300 generations, evaluating 100 candidates over 5 seeds, totaling

150,000 simulation runs and roughly 9.5 hours of wall-clock time. All surrogate-assisted configurations used the same 300-generation span and the same 5-seed evaluation for candidates that received full simulation, giving 49,200 simulation runs in total (9,840 unique candidate evaluations against 30,000 for the baseline).

Performance is measured by final throughput, total simulation count, and wall-clock time. All throughput values reported here are from a 100-simulation robustness test run on the best solution found by each method. The 5-simulation training scores are affected by seed variance; correcting to 100-simulation means reduces this noise and gives a fair basis for comparison.

The primary comparison uses throughput versus cumulative simulation count rather than throughput versus generation index. Surrogate generations use around 164 simulations while full-evaluation generations use 500, so a generation-by-generation plot conflates efficiency with training progress.

Four configurations are compared: (i) Vanilla CMA-ES (full budget baseline), (ii) Surrogate V1 (point-estimate MLP, $\lambda = 0$), (iii) Surrogate V3 (ensemble UCB, $\lambda = 1.0$; V2 was an intermediate configuration not included in the final comparison), and (iv) an ablation with $\lambda = 2.0$. All surrogate configurations use 49,200 simulation runs.

A. Online GGO Extension

To test whether the surrogate framework transfers beyond static edge weights, we apply it to the online GGO setting of Zang et al. [2]. Here the search space is the 4,271-parameter space of a CNN guidance policy. Unlike static edge weight vectors, raw CNN parameters have no direct spatial interpretation, making them significantly harder for the surrogate to model. Each evaluation chains 50 sequential simulator calls, making it roughly $3.8\times$ slower per generation than the offline case (460 s vs. 120 s).

The online vanilla baseline ran for 100 generations with $n_{\text{evals}} = 1$ seed per candidate, totaling 10,000 simulation runs and 13.8 hours of wall-clock time. The surrogate-assisted run used the same three-phase loop with a warmup of 10 generations bootstrapped from vanilla data to avoid redundant re-simulation, giving 3,520 total simulation runs and 4.9 hours of wall-clock time.

B. Robustness Experiments

To assess whether the results generalize beyond the primary configuration, two additional experiments were run using the same hyperparameters as the main offline study. First, the vanilla vs. V3 comparison was repeated on the warehouse map with a different random seed (seed 123) to test seed sensitivity. Second, both methods were applied to a structurally different map (random-32x32, a 32×32 random grid, seed 42) to assess generalization to a different environment. For both additional runs, throughput values are corrected to 100-simulation means.

V. RESULTS

A. Baseline Convergence

The vanilla CMA-ES run improves quickly early on and then slows. Mean population throughput rose from 3.57 at generation 0 to 7.13 at generation 100, 8.05 at generation 200, and roughly 8.29 at generation 300. The curve was clearly flattening by generation 250, gaining only +0.10 over the last 50 generations. The occasional step increases visible in the convergence curve are consistent with the natural dynamics of multi-emitter CMA-ES as individual emitters converge and populations shift.

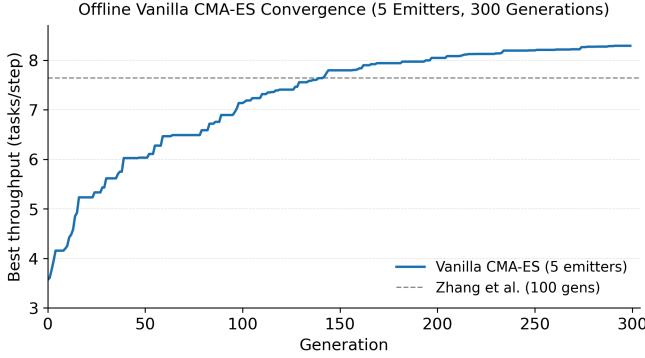


Fig. 2. Baseline CMA-ES convergence. The dashed line shows the throughput reported by Zhang et al. [1] for a 100-generation run, which our 300-generation run surpasses at generation 145.

B. Main Comparison

Fig. 3 shows the sample efficiency comparison. The primary metric is crossover speedup: the number of simulations each method needs to first reach a matched throughput quality of $\tau = 8.23$. Vanilla CMA-ES reaches this threshold at generation 274, corresponding to 137,000 simulations. Surrogate V3 reaches the same threshold at generation 297, using only 49,000 simulations. This gives a crossover speedup of **2.81x**. The V3 run completed in 12,539 seconds (3.5 hours), compared to roughly 9.5 hours for vanilla, a 2.71x wall-clock reduction; the slight excess over the simulation-count ratio reflects fixed surrogate training and inference overhead. Vanilla CMA-ES ran for 300 generations. V3 ran for 400 generations in total; at generation 300 its score is comparable to vanilla’s, and by generation 400 it reaches 8.33, confirming the surrogate does not degrade solution quality over a longer run.

Table I compares all configurations at an equal simulation budget of $\approx 49k$ simulations. At this budget, vanilla CMA-ES has completed roughly 98 generations and reached a corrected throughput of 7.0198. Vanilla reaches only generation 98 because it evaluates 100 candidates per generation over 5 seeds (500 simulation runs); the surrogate screens most candidates and runs roughly 164 real simulations per generation, so the same budget covers the full 300-generation schedule. The generation gap is the mechanism of the quality improvement, not a confound. All three surrogate configurations achieve

substantially higher quality within the same budget. Same budget improvements are reported in Table I.

TABLE I
MATCHED-BUDGET QUALITY COMPARISON ($\approx 49k$ SIMULATIONS EACH).
VANILLA IS EVALUATED AT ITS BEST SOLUTION FOUND BY
GENERATION 98; SURROGATE METHODS RUN THEIR FULL
300-GENERATION SCHEDULE WITHIN THE SAME BUDGET. ALL VALUES
ARE 100-SIMULATION CORRECTED MEANS.

Method	Throughput	95% CI	vs. Vanilla
Vanilla CMA-ES (gen ≈ 98)	7.0198	[7.00, 7.04]	N/A
V1 (point-estimate, $\lambda = 0$)	8.0385	[8.02, 8.05]	+14.5%
Ablation ($\lambda = 2.0$)	7.8594	[7.85, 7.87]	+12.0%
V3 (ensemble, $\lambda = 1.0$)	8.1523	[8.14, 8.17]	+16.1%

C. Robustness Across Seeds and Maps

Table II reports the crossover-point speedup across all four experimental conditions.

Seed 123 yields a 2.92x speedup (Table II), consistent with the seed-42 result and confirming the speedup is not attributable to a fortunate draw. In both warehouse conditions the surrogate’s 100-sim corrected quality also exceeds vanilla’s: +0.084 at seed 42 and +0.033 at seed 123.

On the random-32x32 map the surrogate achieves higher final quality than vanilla, but the crossover speedup is **1.00x**. Because the warmup phase runs full evaluations on both methods using the same seed, the first 20 generations are identical. This map is comparatively simple: both methods reach the quality target well before the warmup phase ends and the surrogate begins screening candidates. On problems that converge this quickly there is no real opportunity for surrogate pre-screening to provide a speedup.

TABLE II
CROSSOVER-POINT SPEEDUP ACROSS ALL EXPERIMENTAL CONDITIONS

Condition	Setting	τ (vanilla)	Speedup
Warehouse seed 42	Offline	8.2205	2.81x
Warehouse seed 123	Offline	8.0176	2.92x
Online GGO	Online	7.433	2.41x
Random-32x32 s42	Offline	3.3642	1.00x[†]

[†]Converges before surrogate phase activates; see text.

D. Surrogate Accuracy in the Live Loop

The mean Spearman ρ in the live surrogate loop was 0.49, below the 0.78–0.94 range from the offline feasibility study. Three factors explain this. First, warmup generations (0–19) produce low- ρ measurements (0.10–0.50) before enough training data has accumulated. Second, as the CMA-ES distribution shifts into new regions of the weight space, ρ temporarily drops to 0.22–0.38 at the first control generation after the shift, because the model has no data for those regions. Third, surrogate generations add only 20 real evaluations each instead of the 100 assumed in the offline study, so training data

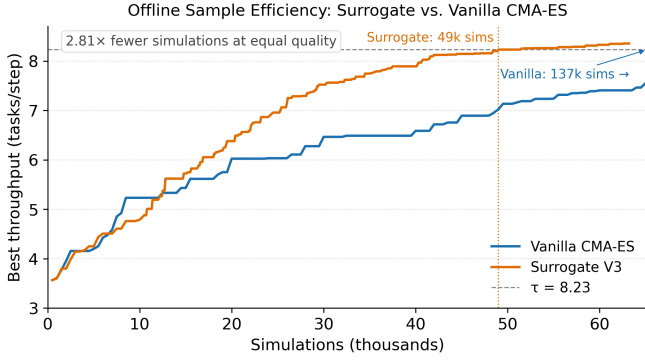


Fig. 3. Sample efficiency: best throughput vs. cumulative simulations. Both methods run 300 generations. The surrogate uses roughly 164 real simulations per generation vs. 500 for vanilla, so slower per-generation progress is expected. Both reach $\tau = 8.23$; the surrogate does so at 49k simulations against 137k for vanilla, a 2.81x crossover speedup.

grows five times more slowly. When warmup and immediate post-shift generations are excluded, ρ generally ranges 0.40–1.0, consistent with the feasibility study results and sufficient for effective rank-based pre-screening in the multi-emitter setting, where the dominant task is identifying which emitter a candidate belongs to rather than fine-grained ranking within a single distribution (see Section V-E).

E. Diagnosing the Source of Surrogate Rank Correlation

The stable-phase rank correlation of 0.40–1.0 confirms that the surrogate provides a useful signal, but it does not reveal what that signal is. A surrogate could achieve high ρ either by learning the fitness landscape (ranking candidates by intrinsic quality within an emitter’s distribution) or by exploiting structural features of the population (identifying which emitter a candidate belongs to). To distinguish these cases, we ran three escalating diagnostic tests on a single-emitter baseline (1 emitter, popsize = 100, 100 generations; same map and seed as the main run), where the second explanation cannot apply by construction.

Test 1: Model capacity (full-leakage). Each model was trained on all 100 candidates from generation g and immediately evaluated on those same 100 candidates. XGBoost achieved $\rho = 0.999$ –1.000 across all test generations, confirming that it can perfectly memorise and rank data it has seen. MLP achieved $\rho = 0.79$ –0.89 under regularisation. Model capacity is *not* the bottleneck.

Test 2: Temporal distribution shift (oracle). Each model was trained on a random 80-candidate subset of generation g and evaluated on the remaining 20 candidates from the *same* generation, eliminating any temporal gap. Both models produced $\rho \approx 0$ (XGBoost mean = +0.03; MLP mean = −0.03), despite perfect in-sample ranking. Distribution shift is *not* the bottleneck.

Test 3: Measurement noise. Given population and seed-variance estimates, the theoretical rank-correlation ceiling is $\rho \leq 0.99$, ruling out measurement noise as the bottleneck [10].

The only remaining explanation is local landscape roughness: the throughput function is non-smooth at the scale of within-generation variation, so knowing the quality of 80 nearby solutions provides essentially no information about 20 others drawn from the same CMA-ES distribution.

This raises the question of why the surrogate works in the five-emitter setting at all. We adapt the intraclass correlation coefficient from psychometrics [11] to quantify population structure in the multi-emitter setting:

$$\text{ICC} = \frac{\sigma_{\text{between}}^2}{\sigma_{\text{between}}^2 + \sigma_{\text{within}}^2}, \quad (4)$$

where $\sigma_{\text{between}}^2$ is the variance of per-emitter mean throughputs and σ_{within}^2 is the mean within-emitter variance. This corresponds to the ICC(1,1) variant of Shrout and Fleiss [11] (one-way random effects, single measures). A high ICC means most population diversity comes from between-emitter differences, making the surrogate’s effective task *cluster identification* rather than fine-grained regression.

Fig. 4 (top panel) shows per-emitter throughput at evolution control generations of the surrogate run: emitter 2 diverges to 8.22 while emitters 0, 1, 3, and 4 plateau near 5.1 from generation 60 onwards, producing the between-emitter variance that drives ICC growth. The bottom panel compares ICC trajectories from the vanilla and surrogate runs.

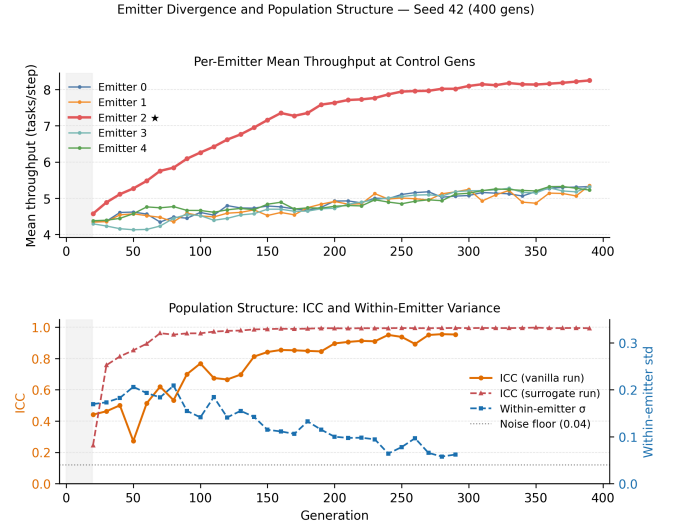


Fig. 4. Top: per-emitter mean throughput at evolution control generations of the surrogate run. Emitter 2 (red) diverges to 8.22 by generation 380 while emitters 0, 1, 3, and 4 plateau near 5.1 from generation 60. Bottom: ICC in the vanilla run (orange solid, left axis) and in the surrogate run (red dashed, left axis), with within-emitter standard deviation (blue, right axis). In the vanilla run ICC grows gradually from 0.44 to 0.95 as emitters diverge naturally; in the surrogate run ICC saturates near 1.0 by generation 40 because UCB screening reinforces emitter 2’s dominance. Within-emitter σ falls toward the measurement noise floor (dotted).

The bottom panel of Fig. 4 shows ICC growing from 0.44 at generation 20 to 0.95 by generation 280 in the vanilla run (orange solid). In the surrogate run (red dashed), ICC saturates near 1.0 by generation 40: UCB screening concentrates simulations on emitter 2, reinforcing its lead and driving

between-emitter divergence far faster than natural evolution alone. Simultaneously, within-emitter σ falls from 0.17 to 0.06, approaching the measurement noise floor of 0.04. At ICC = 0.95, a model that identifies which emitter a candidate belongs to will rank 95% of the population correctly - a coarse task learnable even in a 4,074-dimensional space. In practice this means the surrogate confidently separates emitter 2's candidates from the rest, but with emitters 0, 1, 3, and 4 all producing similar mean throughputs their 80 candidates are largely indistinguishable from one another. This explains why overall ρ sits near 0.60 rather than approaching 1, and why it varies erratically across control generations depending on how those four clusters happen to be sorted by the model at each evaluation. In the single-emitter setting, ICC = 0 by construction; the oracle test confirms that the resulting fine-grained regression task is infeasible due to landscape roughness. To assess whether richer architectures overcome this, we evaluated LargeCNN (five-layer 2-D CNN on the spatial weight tensor), CellTransformer (per-cell attention over all 948 warehouse cells), and a GAT-based GNN operating on the warehouse topology. All three produced $\rho \approx 0$ in oracle mode; the best sliding-window result was LargeCNN at median $\rho = 0.17$ (Table III), well below the $\rho > 0.8$ generally required for reliable screening [3]–[5]. Within-emitter fine-grained ranking therefore appears infeasible independent of model expressivity, and high ICC is a necessary condition for surrogate-assisted CMA-ES.

TABLE III

SPEARMAN ρ BY POPULATION STRUCTURE. FIVE-EMITTER VALUES ARE FROM ALL EVOLUTION CONTROL GENERATIONS OF THE V3 RUN. SINGLE-EMITTER VALUES ARE FROM THE SLIDING-WINDOW TEMPORAL TEST; XGBOOST, MLP, AND SIMPLE CNN WERE EVALUATED ACROSS EIGHT TEST GENERATIONS AND THE REMAINING ARCHITECTURES ACROSS FIVE.

Condition	Median ρ	Range
Five-emitter (control gens)	0.60	0.22–0.84
One-emitter, XGBoost	0.05	0.00–0.12
One-emitter, MLP	0.01	–0.14–0.24
One-emitter, simple CNN	0.14	–0.08–0.33
One-emitter, LargeCNN	0.17	–0.04–0.42
One-emitter, CellTransformer	0.08	0.02–0.21
One-emitter, GNN	0.03	–0.07–0.16

F. Exploration Weight (λ) Ablation

Table I summarises all three configurations. The relationship is non-monotone: $\lambda = 0$ (8.04) and $\lambda = 2.0$ (7.86) both underperform $\lambda = 1.0$ (8.15). At $\lambda = 0$ the surrogate does not evaluate candidates in newly-shifted emitter regions, leaving their CMA-ES updates on stale predictions. At $\lambda = 2.0$ the uncertainty bonus overrides the fitness signal too often, sending low-quality candidates to full simulation. The quality differences across the three settings are modest and based on a single run; the ablation confirms that some exploration weight is preferable to none, but does not establish a precise optimal value.

G. Online GGO Extension

Table IV summarizes the online GGO results. The surrogate reduces the simulation count from 10,000 to 3,520, a 2.84x reduction. A crossover-point analysis gives a precise picture: the vanilla run first reaches a throughput of 7.332 at 8,300 cumulative simulations, while the surrogate reaches the same quality at 3,440 simulations, a crossover speedup of **2.41x**. Both runs are single executions; the 13.8-hour per-run cost makes replication impractical within the scope of this work, so this result should be treated as indicative.

TABLE IV
ONLINE GGO RESULTS (CNN POLICY, 4,271 PARAMETERS)

Method	Throughput	Simulations	Wallclock
Vanilla (online)	7.433	10,000	13.8 h
Surrogate (online)	7.332	3,520	4.9 h

The online surrogate ρ behaved differently from the offline case. Rank correlation was near zero for the first 50 generations (reaching as low as -0.167 at generation 41), then improved sharply above generation 51, stabilizing at 0.55–0.71 for the remainder of the run. This phase transition occurs because raw CNN parameter vectors do not have the smooth, interpretable structure of static edge weight vectors. The MLP surrogate requires approximately 1,800 real evaluations before it can learn useful structure from the parameter space, compared to roughly 1,000 for the offline case. Despite the poor early-phase accuracy, the surrogate still delivered a 2.41x crossover speedup: the weak predictions in the early phase did not derail the search, and the high-accuracy late phase (generation 51–100) drove the final quality improvement from a throughput of 6.21 to 7.33.

VI. DISCUSSION

A. Why Ensemble UCB Closes the Throughput Gap

V3 outperforms V1 because UCB sends candidates in newly-shifted emitter regions to full simulation, keeping the CMA-ES update honest. V1 ranks those candidates at random and corrupts the emitter update, a form of surrogate exploitation that accumulates over multiple distribution shifts. The quality difference between $\lambda = 0$ and $\lambda = 1.0$ is modest (+1.4% on corrected means) and from a single run; both surrogate variants are outperformed on sample efficiency by a single-emitter CMA-ES without any surrogate.

B. Negative Result: Post-Hoc Gradient Refinement

After the main experiments we tested whether gradient ascent through the trained ensemble could improve the best solution found by V3. Starting from the V3 best, we ran 1,000 gradient steps using the uncertainty-penalized score $\mu - \lambda\sigma$ with L2 regularization ($\text{reg} = 10^{-6}$).

The surrogate predicted a gain of +0.95 (from 7.60 to 8.55). The actual simulation showed a decrease of -0.11 from the V3 best (8.23 to 8.12). Two problems compounded. First, the

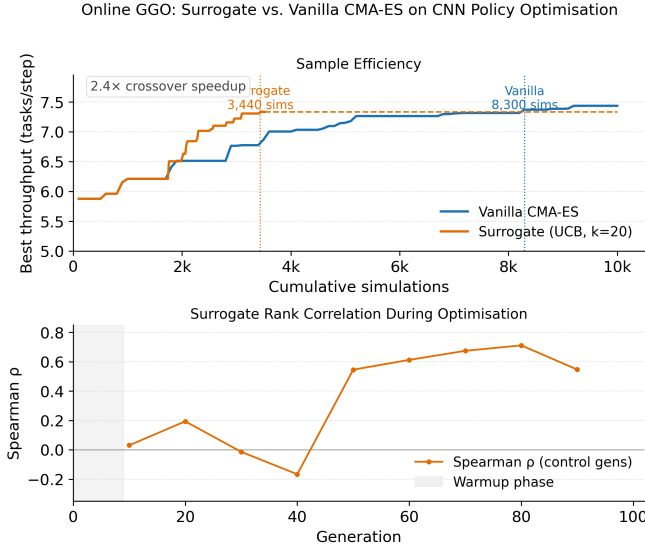


Fig. 5. Online GGO extension. Top: sample efficiency in the online setting; the surrogate reaches the same quality as vanilla at 2.41x fewer simulations. Bottom: surrogate rank correlation over 100 generations, showing a sharp phase transition near generation 51 once roughly 1,800 training samples have accumulated.

surrogate underestimated the V3 best solution by 0.63 units at the starting point, showing poor calibration at the high-quality region. Second, over 1,000 gradient steps the optimizer moved $L_2 = 493$ units through solution space into a region where the model overestimated by +0.44 units. The prediction error changed sign, which is a direct sign of surrogate exploitation.

This confirms why V3 uses the surrogate conservatively. The UCB score selects which of 100 CMA-ES-proposed candidates to evaluate; it never requires the model to be accurate far from the training distribution. Gradient ascent has no such constraint and can be led into regions where model error is large. Direct gradient optimization through this surrogate is not reliable in this setting.

C. Population Structure as a Necessary Condition

The ICC analysis in Section V-E has a direct practical implication: offline feasibility studies conducted on multi-emitter data will systematically overestimate live accuracy whenever ICC in the offline dataset exceeds ICC during deployment. Surrogate-assisted evolutionary algorithms should therefore be benchmarked under population diversity conditions matching deployment, and ICC should be reported alongside accuracy metrics as a predictor of whether surrogate pre-screening will be effective.

VII. CONCLUSION AND FUTURE WORK

The central theoretical finding of this paper is that surrogate pre-screening in multi-emitter CMA-ES is effective because the population’s intraclass correlation coefficient (ICC) grows high as emitters diverge, reducing the surrogate’s task from fine-grained fitness regression to cluster identification. In the

single-emitter case ICC is zero by construction and the surrogate fails regardless of model capacity or data volume, as confirmed by three escalating diagnostic tests. This identifies ICC as a necessary condition for surrogate-assisted CMA-ES, not merely an observed property of the runs.

Under this condition, surrogate-assisted evolutionary strategies substantially reduce the computation required for LMAPF guidance graph optimization. Replacing most simulations with ensemble MLP predictions gives a 2.81x crossover speedup on the primary run: V3 reaches a throughput of $\tau = 8.23$ using 49k simulations, while vanilla CMA-ES requires 137k. A second warehouse seed (123) yields a consistent 2.92x speedup, confirming the result is not seed-dependent. Vanilla ran for 300 generations and V3 for 400; V3’s final score of 8.33 slightly exceeds vanilla’s 8.29, confirming the surrogate does not degrade solution quality. The gradient refinement experiment gives a concrete example of surrogate exploitation: the surrogate predicted a +0.95 gain but simulation measured a -0.11 decrease. The conservative UCB design guards against this failure mode and is recommended as a default for future surrogate work in this setting; the ablation evidence for a specific optimal λ is from a single run and should be treated as indicative.

Robustness experiments on a second random seed and a different map type confirm that the efficiency advantage generalises, with the caveat that problems that converge before the warmup phase ends do not benefit from screening.

We also show that the framework transfers to the online GGO setting of Zang et al. [2], who called for exactly this type of work. Applying the surrogate to a CNN guidance policy (4,271 parameters) reduces simulation count from 10,000 to 3,520, reaching equivalent throughput quality at a 2.41x crossover speedup. The online surrogate exhibits a learning threshold at roughly 1,800 training samples, after which rank correlation rises sharply from near zero to 0.55–0.71. This shows that the surrogate framework generalizes across both static and adaptive guidance representations, and that the required data volume scales with the structural complexity of the search space.

A. Limitations

All offline experiments used a single warehouse map with a fixed agent count and goal distribution; results on varied layouts, densities, or task structures are unknown. The offline comparison covers two random seeds and one additional map type, which is sufficient to rule out a single lucky draw but not to characterize the full variance across environments. The approach requires a multi-emitter configuration to achieve the population diversity that makes surrogate pre-screening feasible (high ICC). To quantify how much of the efficiency gain comes from implicit emitter selection rather than surrogate ranking, we ran a single-emitter baseline: one CMA-ES instance with population size 100 and no surrogate. This baseline reaches the same $\tau = 8.23$ threshold at 41 500 cumulative simulations, a 3.30x reduction relative to the 5-emitter vanilla run, which exceeds Surrogate V3’s 2.81x.

The surrogate therefore does not outperform a simple single-emitter run on this metric, and emitter starvation driven by UCB is the primary mechanism behind the reported speedup. Finally, the online GGO comparison is a single run of each method and should be treated as preliminary given the 13.8-hour per-run cost. Among the architectures evaluated in the single-emitter diagnostic, only LargeCNN offers training times compatible with the live surrogate loop; CellTransformer and the graph neural network require substantially more compute per generation and are not practical as drop-in replacements without further optimisation.

The more productive research directions are alternative genotype representations that expose smoother fitness landscape structure and feature engineering that encodes warehouse graph topology into the surrogate input. If either raises within-emitter rank correlation above the screening threshold, LargeCNN is the most practical starting architecture given its compatible training times.

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